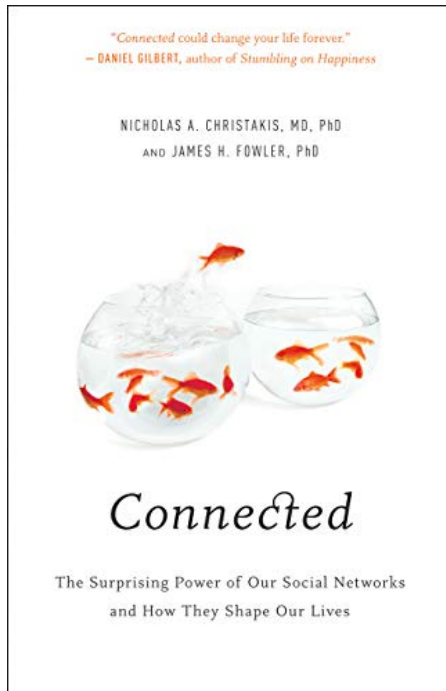


L10: Not Just Viruses Spread On Networks



“Word of mouth” is a powerful influence mechanism and it affects every aspect of our lives.

Think about behavior adoption: Do you like to exercise, eat healthy, party, smoke marijuana? It has been shown again and again that whether someone will adopt such behaviors or not mostly depends on his/her social network. And interestingly, it is not just strong contacts (such as family and close friends) that influence people. Our entire social network, including the crowd of acquaintances and indirect contacts that surround us, also have a strong effect on behavior adoption.

A great reference that summarizes many years of sociology research in this area is the book **“Connected”** by Christakis and Fowler.

Similar influence mechanisms are seen in other aspects of life. For example, in terms of technology adoption, **will you buy an iPhone or an Android device?** Think about your 10 closest people and ask: what phone do they own? You guessed right – your consumer and technology adoption choices are highly influenced by your social network.

Or, in the case of opinion formation: are you a Democrat or a Republican? What do you think about climate change? Abortion? Again, please think about your 10 closest

people and ask the same question. Even though we like to think that we are the absolute owners of our opinions, the reality is that we would probably have very different opinions if we lived in a different social environment.

For centuries, the word-of-mouth mechanism required physical interaction between people. In the last few years, however, online social media such as Facebook, Twitter, or TikTok have provided us with the platform to influence not only the small number of people in our physical proximity – but potentially millions of people around the world.

YouTube or Twitter *“influencers”* today have tens of millions of followers – and it is often the case that their videos or tweets are being liked, commented on, or forwarded by millions of other people. Never before in the history of humankind, we found ourselves in the middle of so many sources of information (*or misinformation*), and never before every single one of us has at least the potential to influence pretty much any other person on the planet.